

Full-Horizon Predictive Battery Thermal Management via Low-Dimensional Link-Costate Optimization

Seunghun Jang¹, Geunyoung Park² and Kyunghwan Choi¹

Abstract—Battery thermal management (BTM) requires long-range anticipation of future cooling demands because slow battery thermal dynamics make effective precooling dependent on distant driving conditions. Conventional model predictive control (MPC), however, must limit its prediction horizon for real-time implementation, which can delay cooling and lead to temperature constraint violations. This paper proposes a full-horizon predictive BTM strategy that incorporates the entire remaining driving information while retaining a low-dimensional online optimization problem. The future driving trajectory is segmented into links and represented by lumped driving parameters, and the BTM optimal control problem is reformulated as a quadratic program over link-wise costates rather than a high-dimensional cooling trajectory. This formulation enables anticipation of distant high-load conditions and continuous updating of the cooling plan over the remaining trip. Simulations with a nonlinear battery model demonstrate effective precooling and reduced temperature constraint violations. Compared with nonlinear model predictive control (NMPC) using a 200-s prediction horizon, the proposed method reduces cooling energy consumption by approximately 3.5% and the maximum temperature constraint violation from 1.419 °C to 0.245 °C. The average and maximum computation times are 20.24 ms and 96.55 ms, respectively, with all optimizations completed within the 1-s control interval.

Index Terms—Battery thermal management, predictive control, full-horizon optimization, link-based optimization, electric vehicles.

I. INTRODUCTION

Battery electric vehicles (BEVs) have been widely developed as an environmentally friendly alternative to conventional internal-combustion-engine vehicles [1], [2]. In BEVs, battery temperature directly affects energy efficiency, safety, performance, and battery lifetime [3]. In particular, excessive heat generation during high-load driving can increase battery temperature and accelerate degradation [4], while maintaining the battery within an appropriate temperature range requires active thermal management. However, battery thermal management (BTM) itself consumes battery energy and can therefore reduce the vehicle driving range [5]. Moreover, because battery thermal dynamics are inherently

slow, cooling actions must anticipate future high-load driving conditions sufficiently in advance. Efficient BTM thus requires long-range optimization of cooling energy while satisfying battery temperature constraints.

BTM has also been formulated as an optimal control problem (OCP) to exploit future driving information. In [6], Pontryagin’s maximum principle (PMP) was applied to optimize battery thermal and energy management, but the approach relied on offline costate estimation. An on-line costate estimation strategy based on PMP was further investigated in [7]. Although such approaches can account for long-range driving information, solving or continuously updating the optimal solution over an entire driving cycle remains challenging for real-time implementation.

Model predictive control (MPC) provides a practical alternative by repeatedly optimizing the cooling input over a finite prediction horizon [5], [8], [9]. However, BTM presents an inherent horizon–computation tradeoff. Because of the slow battery thermal dynamics, a sufficiently long horizon is required to anticipate distant high-load events and initiate precooling in advance. Extending the prediction horizon, however, substantially increases the optimization dimension and computational burden of nonlinear model predictive control (NMPC), whereas a short horizon may identify future cooling demands too late to satisfy temperature constraints. Two-layer and hierarchical MPC strategies have been proposed to incorporate long-horizon predictions using future driving information [5], [8], while adaptive- and multi-horizon strategies have been investigated to reduce the associated computational burden [9]. Nevertheless, incorporating driving information over the entire remaining trip into a computationally tractable online optimization remains challenging.

Recently, a link-based analytical optimization framework was proposed for predictive vehicle energy management [10]. Instead of directly optimizing a high-dimensional time-domain control trajectory, the driving route is divided into links and the optimization is reformulated in terms of link-wise costates, substantially reducing the optimization dimension while retaining long-range driving information. Extending this concept to BTM, however, requires explicit treatment of the slow battery thermal dynamics, cooling-input constraints, and battery-temperature constraints. In particular, the cooling action must be determined sufficiently early to account for high thermal inertia while simultaneously preserving computational feasibility for online implementation.

To address these challenges, this paper develops a full-

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horizon predictive BTM strategy based on link-wise costate optimization. The remaining driving trajectory is represented using lumped driving parameters, and the BTM optimal control problem is reformulated as a low-dimensional quadratic programming (QP) problem whose decision variables are the link-wise costates rather than a long time-domain cooling trajectory. Analytical relations between the link-wise costates, cooling inputs, and battery-temperature evolution enable the cooling-input and temperature constraints to be incorporated directly into the optimization. Consequently, the cooling plan over the entire remaining trip can be repeatedly updated at each control step, allowing distant high-load driving conditions to be anticipated without requiring a long time-discretized NMPC horizon. The main contributions of this study are summarized as follows:

- A full-horizon predictive BTM formulation is developed by reformulating the thermal OCP as a low-dimensional QP over link-wise costates, thereby avoiding direct optimization of a long time-domain cooling trajectory.
- The remaining driving information is compressed into lumped driving parameters, and analytical relations for the cooling input and battery-temperature evolution are derived to incorporate cooling-input and temperature constraints into the optimization.
- The resulting formulation enables the cooling plan over the entire remaining trip to be updated at each control step with computational efficiency suitable for real-time implementation, allowing high-load driving conditions to be anticipated substantially earlier than with finite-horizon NMPC.

II. PRELIMINARIES

A. Battery Electrical and Thermal Models

To solve the OCP for the BTM system, both the electrical and thermal characteristics of the battery are utilized in a prediction model, whose thermal submodel is described as follows [5]:

$$\dot{T}_{\text{bat}}(t) = \frac{1}{m_{\text{bat}}C_{\text{th}}} \left(R_{\text{bat}}I_{\text{bat}}^2(t) + \dot{Q}(t) \right), \quad (1)$$

where T_{bat} , R_{bat} , and I_{bat} are the temperature, internal resistance, and current of the battery pack, respectively, and m_{bat} and C_{th} are its lumped mass and specific heat capacity, respectively. $\dot{Q}$ denotes the heat flow rate required for battery pack cooling and serves as the control input in this model.

The electrical submodel consists of a voltage source connected in series with an internal resistance and is expressed as

$$V_{\text{bat}}(t) = V_{\text{oc}} - R_{\text{bat}}I_{\text{bat}}(t), \quad (2)$$

where V_{bat} and V_{oc} are the battery terminal voltage and open-circuit voltage, respectively. Using the battery power (i.e., $P_{\text{bat}}(t) := V_{\text{bat}}(t)I_{\text{bat}}(t)$) and (2), the battery current is calculated as

$$I_{\text{bat}}(t) = \frac{V_{\text{oc}} - \sqrt{V_{\text{oc}}^2 - 4R_{\text{bat}}P_{\text{bat}}(t)}}{2R_{\text{bat}}}, \quad (3)$$

with

$$P_{\text{bat}}(t) = P_{\text{dem}}(t) + P_{\text{BTM}}(\dot{Q}(t)), \quad (4)$$

where P_{dem} is the demanded traction power and $P_{\text{BTM}}(\dot{Q})$ is the BTM cooling power, which is modeled as a linear function of the cooling input ($\dot{Q}$) [8]:

$$P_{\text{BTM}}(\dot{Q}(t)) = a_c \dot{Q}(t), \quad \dot{Q}(t) \leq 0, \quad a_c < 0, \quad (5)$$

where a_c is the constant coefficient of the cooling system. In this study, the maximum cooling heat flow rate is assumed to be $\dot{Q}_{\text{max}} = 1.5$ kW, with the cooling coefficient selected as $a_c = -3$ based on the SAE J3073 standard [8]. Accordingly, using (3) and (4), the battery state of charge (SOC) is governed by

$$\dot{\text{SOC}}(t) = -\frac{I_{\text{bat}}(t)}{C_{\text{nom}}}, \quad (6)$$

where C_{nom} is the nominal capacity of the battery in coulombs. The battery parameters are obtained from the BEV model environment provided in [11].

B. Optimal Battery Thermal Management Problem

To minimize the total cooling energy consumption over a given time interval ($t \in [t_0, t_N]$), the BTM problem is formulated as

$$\min_{\dot{Q}(t)} \int_{t_0}^{t_N} P_{\text{BTM}}(\dot{Q}(t))dt, \quad (7a)$$

$$\text{subject to (1),} \quad (7b)$$

$$-\dot{Q}_{\text{max}} \leq \dot{Q}(t) \leq 0, \quad (7c)$$

$$T_{\text{bat,min}} \leq T_{\text{bat}}(t) \leq T_{\text{bat,max}}, \quad (7d)$$

$$T_{\text{bat}}(t_0) = T_{\text{bat},0}, \quad (7e)$$

where $T_{\text{bat,min}}$ and $T_{\text{bat,max}}$ are the lower and upper bounds of the battery temperature. $T_{\text{bat},0}$ is the initial battery temperature at time t_0 .

The OCP in (7) is a trajectory optimization problem that determines the cooling input over the entire driving horizon. When the future traction-power demand is known, dynamic programming (DP) can provide an offline benchmark, but its computational cost limits its use in real-time implementation. For online implementation, single-layer nonlinear MPC optimizes a cooling-input sequence over a finite prediction horizon using a time-dependent power-demand trajectory. However, the slow battery thermal dynamics require a sufficiently long horizon to anticipate future cooling needs and initiate precooling [8]. A short horizon may delay cooling and lead to temperature constraint violations [9].

C. Link-Based Segmentation of the Driving Horizon

As discussed in Section II-B, optimizing the input trajectory over the full remaining driving horizon involves a high computational burden. In [10], an analytical approach was proposed to reformulate the optimal energy management problem as a low-dimensional QP over link-wise costates, assuming small state-of-energy (SOE) variations within each link. The reformulated problem incorporates full-horizon

driving information through lumped driving parameters obtained by integration over each link, rather than the full power-demand trajectory over the entire route. This representation can improve robustness to prediction errors.

Likewise, this study extends the link-based approach to full-horizon predictive BTM by assuming small battery temperature and SOC variations within each link. Using stopping intervals identified from the known future driving trajectory, the transportation route is divided into N links with a nominal duration of 30 s for driving links and separate links for stops, where t_i ($i = 1, \dots, N$) denotes the time when the vehicle exits link i (see [10] for details).

III. FULL-HORIZON LINK-COSTATE FORMULATION FOR PREDICTIVE BTM

This section presents a link-based QP formulation of the BTM problem. First, the battery model is simplified and a quadratic objective function is introduced in Section III-A. The link-wise costate parameterization and analytical cooling-input relation are derived in Section III-B. Finally, the temperature change and objective function are expressed in terms of the costates, and the full-horizon QP is formulated in Section III-C.

A. Control-Oriented Model and Objective Reformulation

The nonlinear current model in (3) and thermal dynamics in (1) make the OCP in (7) nonconvex, requiring substantial computational effort to solve using nonlinear programming (NLP). Instead, the OCP is reformulated as a QP by approximating the battery current using a first-order Taylor expansion at zero battery power as follows:

$$I_{\text{bat}}(t) \approx \frac{P_{\text{dem}}(t) + a_c \dot{Q}(t)}{V_{\text{oc}}}, \quad (8)$$

and the corresponding thermal dynamics are expressed as

$$\dot{T}_{\text{bat}}(t) = \underbrace{\frac{R_{\text{bat}} P_{\text{dem}}^2(t) + (2a_c R_{\text{bat}} P_{\text{dem}}(t) + V_{\text{oc}}^2) \dot{Q}(t)}{m_{\text{bat}} C_{\text{th}} V_{\text{oc}}^2}}_{=: f(\dot{Q}(t), P_{\text{dem}}(t))}, \quad (9)$$

where the $a_c^2 R_{\text{bat}} \dot{Q}^2$ term is neglected to obtain the simplified control-oriented model, as $|a_c^2 R_{\text{bat}} \dot{Q}| \ll V_{\text{oc}}^2$ [6]. Since the cooling power is supplied by the battery, cooling operation increases the battery current and the associated ohmic loss. Accordingly, the objective function in (7a) is augmented to account for both the cooling power consumption and the battery resistive loss as

$$l(\dot{Q}(t), P_{\text{dem}}(t)) := P_{\text{BTM}}(\dot{Q}(t)) + R_{\text{bat}} I_{\text{bat}}^2(t). \quad (10)$$

The second term represents the battery ohmic loss associated with the total battery current, including the additional current drawn by the cooling system, and is retained in the link-wise objective derived in Section III-B.

B. Link-Wise Costate Parameterization

With the control-oriented model in (9) and the quadratic objective function in (10), the first step in obtaining a QP formulation is to reformulate the OCP over the entire time interval as a parameter optimization problem in terms of the battery temperature states at link boundaries, using the link-based formulation (see Lemma 1 in [10]):

$$\min_{T_{\text{bat},1}, \dots, T_{\text{bat},N}} \sum_{i=1}^N \min_{\dot{Q}(t)} \int_{t_{i-1}}^{t_i} l(\dot{Q}(t), P_{\text{dem}}(t)) dt, \quad (11a)$$

$$\text{subject to (7c), (7d), (7e), (9)} \quad (11b)$$

$$T_{\text{bat}}(t_i) = T_{\text{bat},i}, \quad i = 1, \dots, N, \quad (11c)$$

where $T_{\text{bat},i}$ is the battery temperature at the end of link i , and N denotes the number of prediction links. To solve the parameter optimization problem in (11), a subproblem is first formulated for each link, whose optimal input trajectory $\dot{Q}^*(t)$ satisfies the following PMP necessary conditions [10]:

$$\min_{\dot{Q}(t)} l(\dot{Q}(t), P_{\text{dem}}(t)) + \lambda(t) f(\dot{Q}(t), P_{\text{dem}}(t)), \quad (12a)$$

$$\text{subject to (7c), (7d),} \quad (12b)$$

$$T_{\text{bat}}(t_{i-1}) = T_{\text{bat},i-1}, T_{\text{bat}}(t_i) = T_{\text{bat},i}, \quad (12c)$$

$$\dot{\lambda}(t) = -\lambda(t) \frac{\partial f(\dot{Q}(t), P_{\text{dem}}(t))}{\partial T_{\text{bat}}}, \quad (12d)$$

where $\lambda(t)$ is the costate associated with the battery thermal dynamics. Assuming small battery temperature and SOC variations within each link, the parameters $R_{\text{bat}}(T_{\text{bat}}, \text{SOC})$ and $V_{\text{oc}}(T_{\text{bat}}, \text{SOC})$ are approximated as constant within that link. Before each QP solve, these parameters are evaluated at nominal temperature and SOC values for each link and held fixed during optimization. The nominal states are initialized from the initial battery state and updated from the preceding prediction, with the current measured state used for the first remaining link. Consequently, $\partial l / \partial T_{\text{bat}} = 0$ and $\partial f / \partial T_{\text{bat}} = 0$ in the approximated model, and (12d) yields $\dot{\lambda}(t) = 0$. Thus, the costate $\lambda(t)$ is replaced by a constant λ_i within each link.

With a constant costate λ_i , the unconstrained stationary input obtained from $\partial H / \partial \dot{Q} = 0$ is expressed as $\dot{Q}_{\text{unc}}(t) = h(P_{\text{dem}}(t), \lambda_i)$. As traction power demand varies, there may be no common range of λ_i for which this unsaturated input satisfies (7c) throughout the link. Input saturation can enforce these input bounds, but introduces piecewise expressions that complicate link integration. To retain analytical relations based on the lumped driving parameters and simple costate bounds, the planned input is restricted to a constant value within each link, i.e., $\dot{Q}(t) = \dot{Q}_i$. For a given λ_i , the constant input is determined by minimizing the Hamiltonian integrated over the link.

With the constant input $\dot{Q}_i$ and costate λ_i within each link,

the Hamiltonian $H(\dot{Q}_i, \lambda_i, t)$ in (12a) is defined as

$$H(\dot{Q}_i, \lambda_i, t) = a_c \dot{Q}_i + \frac{R_{\text{bat}}}{V_{\text{oc}}^2} (P_{\text{dem}}(t) + a_c \dot{Q}_i)^2 + \frac{\lambda_i}{m_{\text{bat}} C_{\text{th}} V_{\text{oc}}^2} \left(R_{\text{bat}} P_{\text{dem}}^2(t) + (2a_c R_{\text{bat}} P_{\text{dem}}(t) + V_{\text{oc}}^2) \dot{Q}_i \right), \quad (13)$$

and integrating (13) over the time interval $[t_{i-1}, t_i]$ yields

$$H_i := \int_{t_{i-1}}^{t_i} H(\dot{Q}_i, \lambda_i, t) dt = a_2 b_1 b_2^2 \Delta t_i \dot{Q}_i^2 + \left(2a_2 b_1 b_2 (1 + a_1 \lambda_i) E_{\text{dem},i} + (b_2 + a_1 \lambda_i) \Delta t_i \right) \dot{Q}_i + a_2 b_1 (1 + a_1 \lambda_i) S_{\text{dem},i}, \quad (14)$$

where $a_1 := 1/(m_{\text{bat}} C_{\text{th}})$, $a_2 := 1/V_{\text{oc}}^2$, $b_1 := R_{\text{bat}}$, and $b_2 := a_c$. The link indices on R_{bat} , V_{oc} , a_2 , and b_1 are omitted for brevity. The lumped driving parameters are given by

$$E_{\text{dem},i} := \int_{t_{i-1}}^{t_i} P_{\text{dem}}(t) dt, \quad (15a)$$

$$S_{\text{dem},i} := \int_{t_{i-1}}^{t_i} P_{\text{dem}}^2(t) dt, \quad (15b)$$

$$\Delta t_i := \int_{t_{i-1}}^{t_i} 1 dt = t_i - t_{i-1}. \quad (15c)$$

The optimal solution for a given λ_i is obtained by solving the stationarity condition $\partial H_i / \partial \dot{Q}_i = 0$ as follows:

$$\dot{Q}_i^* = - \frac{2a_2 b_1 b_2 (1 + a_1 \lambda_i) E_{\text{dem},i} + (b_2 + a_1 \lambda_i) \Delta t_i}{2a_2 b_1 b_2^2 \Delta t_i}. \quad (16)$$

Applying the input bounds in (7c) to (16) yields $\underline{\lambda}_i \leq \lambda_i \leq \bar{\lambda}_i$, where the bounds are given by

$$\underline{\lambda}_i = - \frac{2a_2 b_1 b_2 E_{\text{dem},i} + b_2 \Delta t_i}{a_1 (2a_2 b_1 b_2 E_{\text{dem},i} + \Delta t_i)}, \quad (17a)$$

$$\bar{\lambda}_i = \frac{2a_2 b_1 b_2^2 \Delta t_i \dot{Q}_{\text{max}} - 2a_2 b_1 b_2 E_{\text{dem},i} - b_2 \Delta t_i}{a_1 (2a_2 b_1 b_2 E_{\text{dem},i} + \Delta t_i)}, \quad (17b)$$

where $2a_2 b_1 b_2 E_{\text{dem},i} + \Delta t_i > 0$ holds for all prediction links.

C. Full-Horizon QP Formulation

The second step is to obtain an analytical expression for the temperature change in each link as a function of the costate and lumped driving parameters by integrating the thermal dynamics in (9) over $[t_{i-1}, t_i]$:

$$\int_{t_{i-1}}^{t_i} \dot{T}_{\text{bat}}(t) dt = T_{\text{bat},i} - T_{\text{bat},i-1} = g(\lambda_i, \Theta_i), \quad (18)$$

with

$$g(\lambda_i, \Theta_i) = a_1 a_2 b_1 S_{\text{dem},i} - 2a_1 a_2 b_1 (1 + a_1 \lambda_i) \frac{E_{\text{dem},i}^2}{\Delta t_i} - \frac{a_1}{b_2} (1 + b_2 + 2a_1 \lambda_i) E_{\text{dem},i} - \frac{a_1 (b_2 + a_1 \lambda_i)}{2a_2 b_1 b_2^2} \Delta t_i, \quad (19)$$

where $\Theta_i := [E_{\text{dem},i} \ S_{\text{dem},i} \ \Delta t_i]^\top$ denotes the lumped driving parameter vector for link i .

The last step is to derive an analytical expression for the objective function in (10), integrated over $[t_{i-1}, t_i]$, as a function of the costate and lumped driving parameters:

$$\min_{\dot{Q}_i} \int_{t_{i-1}}^{t_i} \left(a_c \dot{Q}_i + \frac{R_{\text{bat}}}{V_{\text{oc}}^2} (P_{\text{dem}}(t) + a_c \dot{Q}_i)^2 \right) dt. \quad (20)$$

Substituting (16) into (20) yields

$$J(\lambda_i, \Theta_i) = \left(a_1^2 a_2 b_1 \frac{E_{\text{dem},i}^2}{\Delta t_i} + \frac{a_1^2}{b_2} E_{\text{dem},i} + \frac{a_1^2 \Delta t_i}{4a_2 b_1 b_2^2} \right) \lambda_i^2, \quad (21)$$

where the constant terms are omitted without affecting the optimization.

Finally, the QP problem is formulated using (19) and (21), with the costate bounds in (17), as follows:

$$\min_{\Lambda} \Lambda^\top \mathbf{W} \Lambda, \quad (22a)$$

$$\text{subject to } \mathbf{C} \Lambda \leq \mathbf{c}, \quad (22b)$$

$$\underline{\Lambda} \leq \Lambda \leq \bar{\Lambda}, \quad (22c)$$

where

$$\Lambda = [\lambda_1 \ \lambda_2 \ \cdots \ \lambda_N]^\top, \quad \mathbf{W} = \text{diag}([w_1 \ w_2 \ \cdots \ w_N]), \quad w_i = a_1^2 a_2 b_1 \frac{E_{\text{dem},i}^2}{\Delta t_i} + \frac{a_1^2}{b_2} E_{\text{dem},i} + \frac{a_1^2 \Delta t_i}{4a_2 b_1 b_2^2}.$$

The battery temperature constraint matrices are given by

$$\mathbf{C} = [\mathbf{C}_1^\top \ \mathbf{C}_2^\top]^\top, \quad \mathbf{c} = [\mathbf{c}_1^\top \ \mathbf{c}_2^\top]^\top, \quad \mathbf{C}_1 = 2 \begin{bmatrix} w_1 & 0 & \cdots & 0 \\ w_1 & w_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ w_1 & w_2 & \cdots & w_N \end{bmatrix}, \quad \mathbf{C}_2 = -\mathbf{C}_1,$$

$$\mathbf{c}_1 = (T_{\text{bat},0} - T_{\text{bat},\min}) \mathbf{1}_N + \left[\sum_{i=1}^1 d_i \quad \sum_{i=1}^2 d_i \quad \cdots \quad \sum_{i=1}^N d_i \right]^\top,$$

$$\mathbf{c}_2 = (T_{\text{bat},\max} - T_{\text{bat},\min}) \mathbf{1}_N - \mathbf{c}_1,$$

with

$$d_i = a_1 a_2 b_1 S_{\text{dem},i} - 2a_1 a_2 b_1 \frac{E_{\text{dem},i}^2}{\Delta t_i} - \frac{a_1 (1 + b_2)}{b_2} E_{\text{dem},i} - \frac{a_1 \Delta t_i}{2a_2 b_1 b_2},$$

TABLE I
BATTERY PACK PARAMETERS.

Parameter	Value
Pack mass, m_{bat}	150 kg
Specific heat capacity, C_{th}	795 J/(kg · K)
Pack thermal capacity, $m_{\text{bat}}C_{\text{th}}$	119.25 kJ/K
Nominal capacity at 20 °C, C_{nom}	994.5 kC (276.25 Ah)
Open-circuit voltage, $V_{\text{oc}}(T_{\text{bat}}, \text{SOC})$	Lookup table [V]
Internal resistance, $R_{\text{bat}}(T_{\text{bat}}, \text{SOC})$	Lookup table [Ω]

where $\mathbf{1}_N$ denotes the vector of N ones. The costate bounds in (17) are collected as

$$\underline{\Lambda} = [\underline{\lambda}_1 \quad \underline{\lambda}_2 \quad \cdots \quad \underline{\lambda}_N]^\top, \quad \bar{\Lambda} = [\bar{\lambda}_1 \quad \bar{\lambda}_2 \quad \cdots \quad \bar{\lambda}_N]^\top.$$

At each control time step (1 s), the lumped driving parameters for the remaining portion of the current link and the link-wise battery parameters are updated. The QP is then solved over all remaining links using the current measured battery temperature and SOC, with the predicted temperature bounds relaxed by a common nonnegative slack variable penalized linearly in the objective. Only the first input value of the optimized link-wise input trajectory is applied.

Remark 1: Future driving information is required to obtain the lumped driving parameter vectors Θ_i over $[t_0, t_N]$. These parameters can be obtained from a given driving cycle or a speed trajectory predicted using ITS information provided through V2I/V2X communication (e.g., traffic signal states, road conditions, and vehicle information). In this study, future driving information is assumed to be known to validate the proposed formulation for battery thermal management optimization.

IV. VALIDATION

A. Driving Scenario and Comparison Methods

MATLAB simulations are conducted to validate the proposed link-based battery thermal management strategy. All simulations are performed in MATLAB R2025a on a PC equipped with an AMD Ryzen 9 9900X3D CPU and 32 GB of RAM. The target BEV is based on the model environment provided in the IFAC 2026 benchmark problem [11], [12], with the battery parameters listed in Table I.

The driving scenario is obtained from the CarMaker environment and includes 26 intersections defined by traffic light locations. The vehicle passes through the first 20 intersections in the city, enters the highway (1316.5 s), and then returns to the city (1671.6 s) to pass through the remaining 6 intersections before the scenario ends (2149.1 s), as shown in Fig. 1. The driving scenario includes road slopes and both mild and aggressive acceleration and deceleration.

The future vehicle speed trajectory is assumed to be known. Stopping intervals at intersections are identified from this known trajectory and incorporated into the link segmentation. Driving intervals are divided into links of up to 30 s, resulting in a total of 85 links over the given driving scenario.

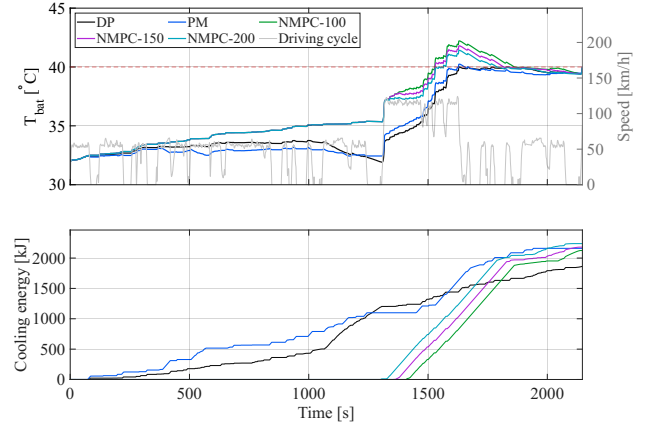


Fig. 1. Battery temperature trajectories and driving scenario (top), and cumulative cooling energy consumption (bottom) for DP, PM, NMPC-100, NMPC-150, and NMPC-200.

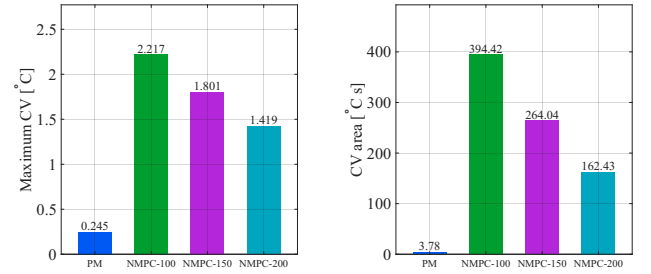


Fig. 2. Maximum battery-temperature constraint violation and cumulative violation area for PM, NMPC-100, NMPC-150, and NMPC-200.

TABLE II
TEMPERATURE CONSTRAINT VIOLATIONS AND COOLING ENERGY CONSUMPTION.

Method	Max. CV [°C]	CV area [°C·s]	Cooling energy [kJ]
DP	0.00	0.00	1861.50 (0.0%)
PM	0.245	3.78	2160.02 (+16.0%)
NMPC-100	2.217	394.42	2124.35 (+14.1%)
NMPC-150	1.801	264.04	2179.69 (+17.1%)
NMPC-200	1.419	162.43	2238.02 (+20.2%)

Battery heat generation increases due to the high driving load after highway entry. Thus, given the slow battery thermal dynamics, future cooling demand should be anticipated and the battery temperature sufficiently lowered before highway entry to reduce thermal stress, preserve battery performance, and extend battery life [13]. In this study, the battery temperature bounds are set to $T_{\text{bat},\text{min}} = 25$ °C and $T_{\text{bat},\text{max}} = 40$ °C throughout the driving scenario, with a control time step of 1 s.

B. Thermal Management and Energy Performance

The proposed method (PM) is compared with DP and NMPC, where the objective in (7a) is replaced by (10), using the nonlinear battery model for both DP and NMPC. All methods use the same objective function, initial battery temperature, and temperature and input bounds, with an identical slack penalty to handle infeasible temperature constraints.

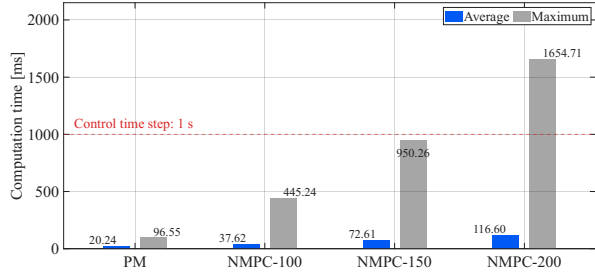


Fig. 3. Average and maximum optimization time of PM, NMPC-100, NMPC-150, and NMPC-200.

The QP problem for PM is solved using the MATLAB `quadprog` function with the interior-point-convex algorithm, while the nonlinear optimization problems for NMPC are solved using the `fmincon` function with the sequential quadratic programming (SQP) algorithm.

Figure 1 compares the battery temperature trajectories (top) and cumulative cooling energy consumption (bottom) of PM [—], DP [—], and NMPC with prediction horizons of $N_p = 100, 150$, and 200 s, denoted by NMPC-100 [—], NMPC-150 [—], and NMPC-200 [—], respectively. The driving scenario is shown in gray [—]. The cumulative cooling energy is calculated as the time integral of the BTM power P_{BTM} .

The NMPC controllers determine the cooling input based on the known future load within their respective finite prediction horizons. However, because of the slow battery thermal dynamics, these horizons limit how early the controllers can anticipate the high thermal load associated with highway driving. As a result, the cooling actions are initiated relatively late, leading to insufficient precooling before highway entry and subsequent temperature constraint violations.

In contrast, PM incorporates the traction-power demand over the full remaining driving horizon through the lumped driving parameters. Consequently, the controller identifies the upcoming high-load highway segment sufficiently early and initiates precooling before highway entry. This anticipatory behavior is reflected by the earlier increase in cumulative cooling energy in Fig. 1, resulting in a temperature trajectory closer to that of DP and substantially smaller temperature constraint violations.

Figure 2 compares the maximum temperature constraint violation and the cumulative violation area. PM limits the maximum violation to 0.245°C with a violation area of $3.78^\circ\text{C}\cdot\text{s}$, whereas NMPC-100, NMPC-150, and NMPC-200 exhibit maximum violations of 2.217°C , 1.801°C , and 1.419°C , respectively. The corresponding violation areas are $394.42^\circ\text{C}\cdot\text{s}$, $264.04^\circ\text{C}\cdot\text{s}$, and $162.43^\circ\text{C}\cdot\text{s}$. These results indicate that extending the NMPC prediction horizon improves anticipatory cooling, but the considered finite horizons remain insufficient to fully capture the distant cooling demand associated with the high-load driving segment.

Table II summarizes the temperature-constraint violations and cooling-energy consumption. PM consumes 2160.02 kJ of cooling energy, which is 16.0% higher than the offline DP

benchmark. Nevertheless, compared with NMPC-200, PM reduces the cooling-energy consumption by approximately 3.5% while decreasing the maximum temperature constraint violation from 1.419°C to 0.245°C . Thus, the improved constraint handling of PM is not achieved by simply increasing the cooling-energy consumption relative to the longest-horizon NMPC case. Rather, the full-horizon information enables the cooling energy to be allocated earlier and more effectively through anticipatory precooling.

C. Computational Performance

Figure 3 compares the optimization times of PM and the three NMPC controllers. PM achieves an average optimization time of 20.24 ms and a maximum optimization time of 96.55 ms, with all optimizations completed well within the 1 -s control interval. In contrast, the computational burden of NMPC increases with the prediction horizon. In particular, NMPC-200 requires a maximum optimization time of 1654.71 ms, exceeding the control interval.

It should also be noted that the longest NMPC prediction horizon considered in this study is 200 s, whereas PM incorporates driving information over the full remaining trip. The considered driving scenario has a total duration of approximately 2149 s and is represented by up to 85 links. Extending nonlinear MPC to a comparable time horizon would require a substantially larger time-domain optimization problem, whereas the proposed formulation represents the horizon using link-wise costates as optimization variables. These results demonstrate that reformulating the thermal-management problem in terms of link-wise costates enables full-horizon predictive thermal management to be repeatedly updated within the 1 -s control interval while retaining the long-range information required for anticipatory cooling.

V. CONCLUSIONS

This study developed a full-horizon predictive BTM strategy that enables long-range anticipation of thermal loads while retaining computational feasibility for real-time implementation. Instead of directly optimizing a long time-domain cooling trajectory, the proposed method represents the remaining driving horizon through lumped driving parameters and reformulates the thermal OCP as a low-dimensional QP over link-wise costates. This formulation allows the cooling plan over the full remaining driving horizon to be updated at each control step.

Simulation results with a nonlinear battery model demonstrated that the proposed method anticipates upcoming high-load driving and initiates precooling before highway entry, resulting in substantially improved temperature-constraint handling. Compared with NMPC-200, the proposed method reduced the maximum temperature constraint violation from 1.419°C to 0.245°C while reducing cooling energy consumption by approximately 3.5% . The average and maximum optimization times were 20.24 ms and 96.55 ms, respectively, and all optimizations were completed within

the 1-s control interval. Although the cooling energy consumption remained 16.0% higher than that of the offline DP benchmark, the proposed method achieved a favorable balance between long-range thermal anticipation, energy efficiency, and online computational feasibility. The present study assumes that full-horizon driving information is available and enforces temperature constraints only at link boundaries.

Future work will address uncertainty in driving prediction and model approximation, as well as temperature constraint satisfaction within individual links.

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